

C.3 Effectiveness on Zero-shot and One-shot Inference

When facing zero-shot and one-shot inference tasks, H₂O can successfully mitigate the memory requirements of the KV cache by up to $5\times$ under both one/zero-shot inference across different tasks, achieving matching performance as the model with full KV cache while the local method fails. However, unlike 5-shot, we also found that some tasks are more difficult, and the model requires more information to generate the correct content, resulting in a higher KV cache budget (30-40%). This might be expected since 0/1 shots in our benchmarks have shorter sequence lengths ranging from 100 to 300.

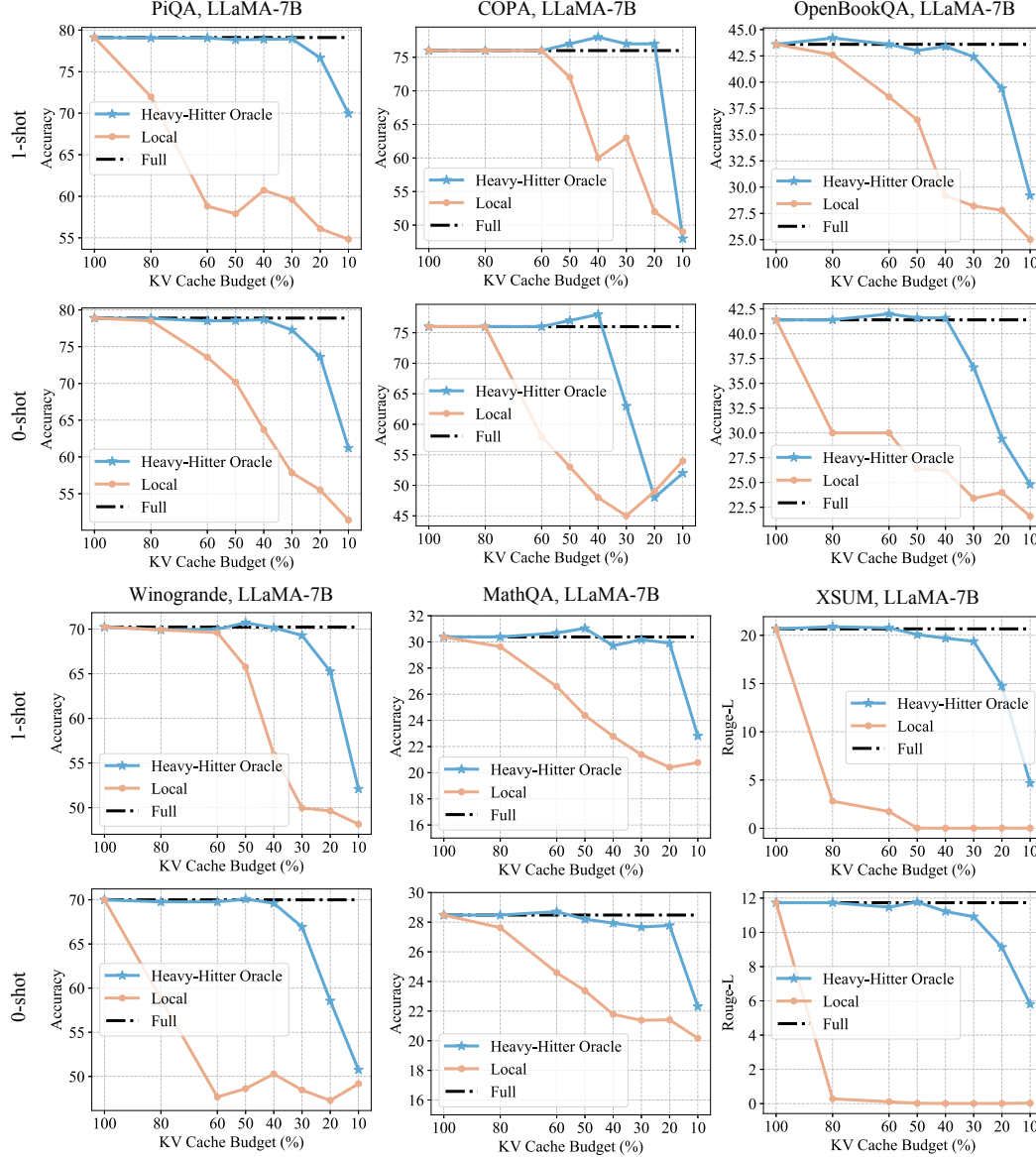


Figure 8: Comparison results on zero-shot and one-shot inference.

C.4 Comparison with StreamingLLM

Recent works, such as StreamLLM [52] and LM-Infinite [53], have shown promising potential in enabling Language Models (LLMs) to handle input of infinite length. They achieve this by only retaining the initial tokens and a limited local context. However, this approach may pose challenges for certain tasks where vital information lies within the middle of the input and would be lost using this strategy. We investigate it through two specific types of tasks:

Multi-document question answering [95]: In this task, each test sample comprises ten documents, followed by a question. The key information to answer this question is stored in one of the documents. We rearranged the crucial document’s position and found that the eviction strategy in StreamLLM or LM-Infinite can not perform well when the key document has been dropped.

Text Summarization Task (XSUM and CNN-DailyMail): Text summarization tasks require models to generate concise summaries of lengthy texts. Effective summarization demands a comprehensive understanding of the entire document, making it challenging when crucial information is dispersed throughout the input. In particular, summarization often relies on long-context attention, and critical information may not be effectively captured within the limited local tokens.

The results are reported in Figure 9, illustrating a consistent decline in the performance of StreamLLM. Since StreamLLM will always maintain the first few tokens as well as the local tokens, regardless of various input content, such a strategy will inevitably result in the loss of crucial information and subsequently lead to a decrease in performance. In contrast, our H₂O delivers markedly superior performance. Also, with H₂O, The model can successfully stream to four million tokens.

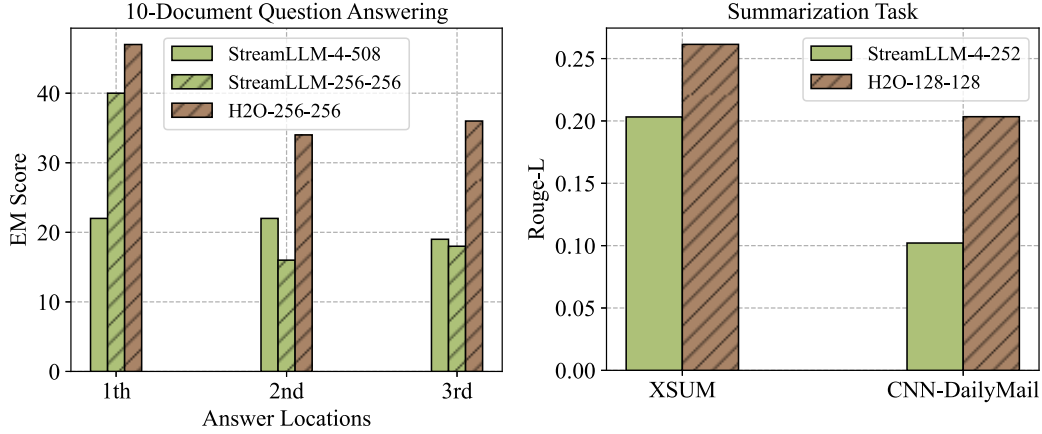


Figure 9: Comparison results of StreamLLM [52] and our H₂O on generation tasks. The number in each method represents the KV Cache budget of the start/heavy-hitter tokens and the local tokens, respectively. For example, H2O-256-256 means maintaining 256 Heavy-Hitters and 256 local tokens.

C.5 Enhancing the "Top-K" Baseline

We find H₂ can further enhance another strong baseline with a "Top-K" strategy. The results are reported in Table 8. After combining with H₂, the "Top-K" method achieves an extra improvement with up to 2.00% accuracy across 4 different tasks.

Table 8: Results of the "Top-K" method w. or w.o. H₂. Experiments are conducted with OPT-30B with 20% KV cache budget.

Models	COPA	OpenBookQA	PiQA	Winogrande
Full	85.00	43.20	78.51	70.24
TopK w.o. H ₂	80.00	41.40	76.96	65.35
TopK w. H ₂	82.00	42.80	77.96	66.48

C.6 Separate Effects of Heavy-Hitter and Local Tokens

Table 9 demonstrates the separate effects of Heavy-Hitters and the local tokens. And we can observe Heavy-Hitters contribute more to maintaining the performance of models.

C.7 Performance of Inference with Different Number of Shots.

Table 10 demonstrates our H₂O achieves consistent improvements across different number of shots during inference and maintains the full model’s performance with 20% memory budget.

Table 9: Ablation study of H₂O across different tasks.

Tasks	Models	Full	w. Local	w. H ₂	w. Local + H ₂
PiQA	OPT-13B	77.37	54.62	76.12	77.26
	OPT-30B	78.51	55.82	67.25	78.45
OpenBookQA	OPT-13B	41.40	25.60	30.40	41.20
	OPT-30B	43.20	25.20	26.60	43.00
MathQA	OPT-13B	26.67	22.04	23.82	26.93
	OPT-30B	26.23	20.87	21.98	26.87
Winogrande	OPT-13B	68.59	49.96	51.85	67.32
	OPT-30B	70.24	49.17	47.36	69.06

Table 10: Results under different sequence length of OPT-30B with 20% KV cache budget.

Tasks	Methods	5-shots		10-shots	
		OPT-30B	OPT-66B	OPT-30B	OPT-66B
OpenBookQA	Full	43.20	44.40	43.00	44.80
	Local	25.20	30.60	26.60	38.80
	H ₂ O	43.00	44.20	42.80	44.80
COPA	Full	85.00	83.00	86.00	85.00
	Local	48.00	59.00	60.00	76.00
	H ₂ O	84.00	82.00	85.00	86.00
MathQA	Full	26.23	27.87	26.67	27.00
	Local	20.87	25.49	21.11	23.08
	H ₂ O	26.87	27.67	26.47	27.30

C.8 Heavy-Hitter in Attention Blocks

The distribution of accumulated attention scores of all the tokens within attentions blocks is illustrated in Figure 10. We can observe that H₂ broadly exists in each layer.

C.9 Comparision with SpAtten

Compared with SpAtten [32], the main differences of H₂O are i) They accumulate attention scores across attention heads and layers, while in our algorithm, each token can be kept or evicted independently across heads and layers, providing more flexibility for selecting critical KV embeddings; ii) We also use KV of the most recent tokens during generation and demonstrate that such H₂ tokens can effectively enhance other sparse attention strategies; iii) we formulate the KV cache eviction as a dynamic submodular problem and prove a theoretical guarantee (under mild assumptions) for our novel algorithms. Moreover, we also provide a quantitative comparison with SpAtten, and the results are reported in Table 11.

Table 11: Comparison between SpAtten [32] and H₂O across various tasks with OPT-30B.

Models	COPA	OpenBookQA	PiQA
Full	85.00	43.20	78.51
SpAtten	82.00	41.90	77.06
H ₂ O	84.00	43.00	78.45

C.10 Heavy-Hitter in MLP Blocks

Besides the attention blocks, the presence of Heavy-Hitters (H₂) is observed within the MLP blocks of LLMs. We utilize the Wiki-Text-103 dataset as the input and record the activated frequency of neurons in the hidden layer of MLP blocks. As depicted in Figure 11, the activated frequency of neurons follows a power-law distribution, wherein a small number of neurons are activated by nearly all input tokens (with a 100% frequency) while the majority of other neurons are rarely activated.

Subsequently, a thorough examination of various characteristics pertaining to H₂ in MLP blocks is conducted, encompassing the following aspects: (1) The elimination of H₂ leads to a substantial decline in performance, although such degradation can be easily recovered even with a mere 1% of the training data; (2) H₂ exhibits a significant degree of overlap across different type of input content; (3) The emergence of H₂ occurs early in the training process, thus exhibiting an "early-bird" characteristic, and their positions undergo gradual changes during subsequent training phases.